

A NOTE ON A UNIVERSAL ROUGH PARITY BIAS IN THE POLYNOMIAL GOLDBACH PROBLEM

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ABSTRACT. Fix an integer $n \geq 3$ and an odd prime power q . For a monic polynomial $F \in \mathbb{F}_q[t]$ of degree n , consider decompositions $F = g + h$ in which g is monic irreducible of degree $n - 1$ and h is squarefree with no irreducible factor of degree at most y . We prove, in the large- q limit, that the number of such decompositions for which h has an odd number of irreducible factors exceeds the number for which it has an even number of factors, uniformly in F and in every threshold $1 \leq y < n$. The main term is explicit. The combinatorial input is the exact identity

$$\frac{1}{n!} \sum_{\substack{\sigma \in S_n \\ \text{all cycles of } \sigma \text{ have length} > y}} (-1)^{\text{cyc}(\sigma)} = -\frac{p_y(n-y-1)}{n},$$

where $p_y(m)$ is the probability that a random permutation in S_m has all cycles of length at most y . The arithmetic input is a class-function refinement implicit in the Bender–Pollack proof of the polynomial Goldbach theorem. We also give a Liouville-weighted formulation, a prescribed source factorization-type variant, and a homological-sieve interpretation.

1. INTRODUCTION

Let $\mathcal{M}_m(q)$ and $\mathcal{I}_m(q)$ denote, respectively, the monic polynomials and the monic irreducible polynomials of degree m in $\mathbb{F}_q[t]$. Bender and Pollack [2] proved an asymptotic formula, in the limit of fixed degree and growing odd q , for the number of decompositions

$$F = g + h, \quad g \in \mathcal{I}_{n-1}(q), \quad h \in \mathcal{I}_n(q),$$

uniformly over monic F of degree n . Their proof constructs a geometric Galois extension with group $S_{n-1} \times S_n$ and applies an explicit Chebotarev theorem.

Here we keep the first summand irreducible but replace irreducibility of the second summand by a roughness condition. For a nonconstant polynomial h , write

$$d^-(h) = \min_{P|h} \deg P,$$

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where P runs over monic irreducible divisors. For $1 \leq y < n$, define

$$N_{F,y}^{\text{odd}} = \# \left\{ g \in \mathcal{I}_{n-1}(q) : \begin{array}{l} F - g \text{ is squarefree, } d^-(F - g) > y, \\ \omega(F - g) \text{ is odd} \end{array} \right\},$$

$$N_{F,y}^{\text{even}} = \# \left\{ g \in \mathcal{I}_{n-1}(q) : \begin{array}{l} F - g \text{ is squarefree, } d^-(F - g) > y, \\ \omega(F - g) \text{ is even} \end{array} \right\},$$

where $\omega(h)$ is the number of distinct irreducible factors of h .

For $m \geq 0$, let

$$(1.1) \quad p_y(m) = \frac{1}{m!} \# \{ \tau \in S_m : \text{every cycle has length at most } y \}.$$

In particular, $p_y(0) = 1$. Our main result is the following.

Theorem 1.1 (Universal rough parity bias). *Fix $n \geq 3$. As $q \rightarrow \infty$ through odd prime powers, uniformly for all monic $F \in \mathbb{F}_q[t]$ of degree n and all integers $1 \leq y < n$,*

$$(1.2) \quad N_{F,y}^{\text{odd}} - N_{F,y}^{\text{even}} = \frac{p_y(n-y-1)}{n(n-1)} q^{n-1} + O_n(q^{n-3/2} + q^{n-2}).$$

Consequently, there is a constant $q_0(n)$ such that, for every odd $q \geq q_0(n)$, all of these differences are positive simultaneously. More precisely,

$$(1.3) \quad N_{F,y}^{\text{odd}} - N_{F,y}^{\text{even}} \geq \frac{q^{n-1}}{2n!}.$$

The phenomenon already occurs at the weakest nontrivial threshold $y = 1$: excluding only linear factors produces a positive main-term gap $q^{n-1}/n!$. At $y = \lfloor n/3 \rfloor$, the target has at most two irreducible factors, so [theorem 1.1](#) says that irreducible targets outnumber rough semiprime targets. At $y \geq n/2$, the target can only be irreducible, and the theorem reduces to the usual polynomial Goldbach main term.

The equidistribution statement used below is a routine class-function consequence of the proof in [\[2\]](#); it is not a new monodromy theorem. The signed permutation identity is elementary. The point of this note is that their combination yields a uniform hierarchy of strict arithmetic inequalities, apparently not recorded explicitly in the literature. Related work on factorization statistics and correlations of factorization functions over finite fields includes [\[3, 5\]](#). Unsigned permutations without short cycles have been studied extensively; see [\[6\]](#).

2. A SIGNED CYCLE IDENTITY

For $\sigma \in S_n$, write $\text{cyc}(\sigma)$ for its number of cycles. Define the signed density

$$(2.1) \quad A_{n,y} = \frac{1}{n!} \sum_{\substack{\sigma \in S_n \\ \text{every cycle of } \sigma \text{ has length } > y}} (-1)^{\text{cyc}(\sigma)}.$$

Proposition 2.1 (Exact signed density). *For every pair of integers $n > y \geq 1$,*

$$(2.2) \quad A_{n,y} = p_y(n) - p_y(n-1) = -\frac{p_y(n-y-1)}{n} < 0.$$

Proof. The exponential formula for permutations, with weight -1 assigned to every cycle, gives the formal power-series identity

$$\begin{aligned} \sum_{n \geq 0} A_{n,y} z^n &= \exp \left(- \sum_{j > y} \frac{z^j}{j} \right) \\ (2.3) \qquad \qquad &= (1 - z) \exp \left(\sum_{j=1}^y \frac{z^j}{j} \right). \end{aligned}$$

The ordinary generating function of the probabilities in (1.1) is

$$(2.4) \qquad P_y(z) := \exp \left(\sum_{j=1}^y \frac{z^j}{j} \right) = \sum_{m \geq 0} p_y(m) z^m;$$

see, for example, [4, Chapter II]. Comparing coefficients in (2.3) proves $A_{n,y} = p_y(n) - p_y(n-1)$.

Differentiating (2.4) gives

$$P'_y(z) = (1 + z + \cdots + z^{y-1}) P_y(z),$$

and hence

$$(2.5) \qquad m p_y(m) = \sum_{j=1}^{\min(y,m)} p_y(m-j).$$

For $n > y$, subtracting the instance $m = n-1$ of (2.5) from the instance $m = n$ yields

$$n p_y(n) - (n-1) p_y(n-1) = p_y(n-1) - p_y(n-y-1).$$

Rearranging proves the second equality in (2.2). Finally, $p_y(n-y-1) > 0$ because the identity permutation is counted. $\square$

Corollary 2.2 (Uniform combinatorial gap). *For every $1 \leq y < n$,*

$$(2.6) \qquad \frac{p_y(n-y-1)}{n(n-1)} \geq \frac{1}{n!}.$$

Equality holds when $y = 1$.

Proof. Put $m = n-y-1 \leq n-2$. Since the identity is one of the permutations counted by $p_y(m)$,

$$p_y(m) \geq \frac{1}{m!} \geq \frac{1}{(n-2)!}.$$

For $y = 1$, only the identity has all cycles of length at most one, so $p_1(n-2) = 1/(n-2)!$. $\square$

Remark 2.3. Without a roughness condition one has

$$\frac{1}{n!} \sum_{\sigma \in S_n} (-1)^{\text{cyc}(\sigma)} = 0 \quad (n \geq 2).$$

Thus [proposition 2.1](#) says that removing only the fixed-point stratum already breaks the symmetric-group version of the parity cancellation.

3. THE JOINT FACTORIZATION INPUT

If $f \in \mathcal{M}_d(q)$ is squarefree, its factor degrees determine a conjugacy class in S_d . We write σ_f for an arbitrary representative; a class function evaluated at σ_f is therefore well defined. For a class function a on S_d , set

$$\text{Av}_{S_d}(a) = \frac{1}{d!} \sum_{\sigma \in S_d} a(\sigma).$$

Proposition 3.1 (Class-function form of the Bender–Pollack input). *Fix $n \geq 3$, and let a and b be class functions on S_{n-1} and S_n , respectively. For every monic $F \in \mathbb{F}_q[t]$ of degree n and every odd q ,*

$$(3.1) \quad \sum_{\substack{g \in \mathcal{M}_{n-1}(q) \\ g(F-g) \text{ squarefree}}} a(\sigma_g) b(\sigma_{F-g}) = q^{n-1} \text{Av}_{S_{n-1}}(a) \text{Av}_{S_n}(b) + O_{n,a,b}(q^{n-3/2} + q^{n-2}).$$

The estimate is uniform in F . If $|a|, |b| \leq 1$, the implied constant may be taken to depend only on n .

Proof. It is enough to prove the assertion when a and b are indicators of conjugacy classes $C_1 \subset S_{n-1}$ and $C_2 \subset S_n$. In the proof of [2, Lemma 2.2], Bender and Pollack construct, for every nonexceptional auxiliary polynomial, two splitting fields whose Galois groups are S_{n-1} and S_n . They prove that the compositum is geometric and that the natural map from its Galois group to $S_{n-1} \times S_n$ is an isomorphism. Irreducibility is then detected by choosing C_1 and C_2 to be the two long-cycle classes.

Nothing in the geometric construction, the exceptional-set estimate, or the genus bound depends on that choice of conjugacy classes. Replacing the two long-cycle classes by arbitrary C_1, C_2 in their Chebotarev step gives density

$$\frac{|C_1|}{(n-1)!} \frac{|C_2|}{n!}$$

with the same square-root error. On every unramified fiber, the usual Frobenius–factorization dictionary identifies these two conjugacy classes with σ_g and σ_{F-g} . Their double-counting argument then gives (3.1); ramified and exceptional parameters contribute the $O_n(q^{n-2})$ term. Finally, every class function is a finite linear combination of conjugacy-class indicators. Since the number of conjugacy classes depends only on n , the asserted uniformity for bounded class functions follows. $\square$

Remark 3.2. Taking a and b to be the indicators of the $(n-1)$ -cycle and n -cycle classes gives the main term $q^{n-1}/(n(n-1))$ of the Bender–Pollack Goldbach theorem.

4. PROOF OF THE PARITY-BIAS THEOREM

Define the class function on S_n

$$(4.1) \quad \phi_{n,y}(\sigma) = (-1)^{\text{cyc}(\sigma)} \mathbf{1}_{\{\text{every cycle of } \sigma \text{ has length } > y\}}.$$

By [proposition 2.1](#),

$$(4.2) \quad \text{Av}_{S_n}(\phi_{n,y}) = -\frac{p_y(n-y-1)}{n}.$$

Proof of theorem 1.1. Apply [proposition 3.1](#) with a equal to the indicator of the $(n-1)$ -cycle class and $b = \phi_{n,y}$. The average of a is $1/(n-1)$. Moreover, if $g \in \mathcal{I}_{n-1}(q)$ and $F-g$ is squarefree and y -rough, then g and $F-g$ are automatically coprime: otherwise $g \mid F-g$, so $(F-g)/g$ is linear, contradicting $d^-(F-g) > y \geq 1$. Thus the left-hand side of [\(3.1\)](#) is exactly

$$N_{F,y}^{\text{even}} - N_{F,y}^{\text{odd}}.$$

Equation [\(1.2\)](#) now follows from [\(4.2\)](#).

The error is uniform in y , because the functions $\phi_{n,y}$ are bounded by one and there are only $n-1$ possible thresholds. By [corollary 2.2](#), the main-term coefficient in [\(1.2\)](#) is at least $1/n!$. Choosing $q_0(n)$ so that the error is at most $q^{n-1}/(2n!)$ proves [\(1.3\)](#). $\square$

The same argument allows a prescribed factorization type on the source side.

Proposition 4.1 (Prescribed source types). *Let $B \subseteq S_{n-1}$ be conjugacy invariant. Among $g \in \mathcal{M}_{n-1}(q)$ for which $g(F-g)$ is squarefree, $\sigma_g \in B$, and $d^-(F-g) > y$, let $N_{F,B,y}^{\text{odd}}$ and $N_{F,B,y}^{\text{even}}$ count those for which $\omega(F-g)$ is odd and even, respectively. Then, uniformly in F , y , and B ,*

$$(4.3) \quad N_{F,B,y}^{\text{odd}} - N_{F,B,y}^{\text{even}} = \frac{|B|}{(n-1)!} \frac{p_y(n-y-1)}{n} q^{n-1} + O_n(q^{n-3/2} + q^{n-2}).$$

Proof. Take $a = \mathbf{1}_B$ and $b = \phi_{n,y}$ in [proposition 3.1](#), and negate the resulting signed sum. $\square$

We next remove the squarefree condition by using the polynomial Liouville function

$$\lambda_q(h) = (-1)^{\Omega(h)},$$

where irreducible factors are counted with multiplicity.

Lemma 4.2 (The discriminant boundary). *Uniformly for monic F of degree n ,*

$$\#\{g \in \mathcal{M}_{n-1}(q) : F-g \text{ is not squarefree}\} = O_n(q^{n-2}).$$

Proof. Write the coefficient of t^{n-1} in $F-g$ as c . As g varies in $\mathcal{M}_{n-1}(q)$, the polynomial $F-g$ ranges over the affine hyperplane

$$t^n + ct^{n-1} + a_{n-2}t^{n-2} + \cdots + a_0.$$

The discriminant restricts to a nonzero polynomial on this hyperplane. Indeed, over an algebraic closure one can choose n distinct roots whose sum is $-c$. The restricted discriminant has degree $2n-2$, so the Schwartz–Zippel bound gives $O_n(q^{n-2})$ zeros. $\square$

Corollary 4.3 (Rough Liouville trace). *Fix $n \geq 3$. Uniformly for monic F of degree n and $1 \leq y < n$,*

$$(4.4) \quad \mathcal{L}_{F,y} := \sum_{\substack{g \in \mathcal{I}_{n-1}(q) \\ d^-(F-g) > y}} \lambda_q(F-g) = -\frac{p_y(n-y-1)}{n(n-1)} q^{n-1} + O_n(q^{n-3/2} + q^{n-2}).$$

In particular, $\mathcal{L}_{F,y} < 0$ for all sufficiently large odd q , uniformly in F and y .

Proof. On the squarefree locus, $\lambda_q(F - g) = (-1)^{\omega(F-g)}$, and the claim is the signed form of [theorem 1.1](#). By [lemma 4.2](#), the omitted nonsquarefree locus contributes $O_n(q^{n-2})$. $\square$

5. SPECIAL THRESHOLDS

The coefficient in [theorem 1.1](#) has transparent forms in several ranges.

5.1. Excluding only linear factors. For $y = 1$, the only permutation all of whose cycles have length at most one is the identity. Hence

$$p_1(n-2) = \frac{1}{(n-2)!}, \quad N_{F,1}^{\text{odd}} - N_{F,1}^{\text{even}} = \frac{q^{n-1}}{n!} + O_n(q^{n-3/2}).$$

Thus the smallest possible roughness restriction already creates a strict parity bias.

5.2. The one-factor/two-factor threshold. Let $y = \lfloor n/3 \rfloor$ and put

$$\beta_n = \sum_{n/3 < j < 2n/3} \frac{1}{j}.$$

For $m = n - y - 1$, two cycles of length greater than y cannot both occur in S_m . A random permutation in S_m contains a j -cycle with probability $1/j$, and these events are disjoint for $j > y$. It follows that

$$(5.1) \quad p_y(n - y - 1) = 1 - \beta_n.$$

Since an n -degree y -rough polynomial has at most two irreducible factors, [theorem 1.1](#) becomes

$$\begin{aligned} & \#\{\text{irreducible targets}\} - \#\{y\text{-rough semiprime targets}\} \\ &= \frac{1 - \beta_n}{n(n-1)} q^{n-1} + O_n(q^{n-3/2}). \end{aligned}$$

5.3. The full hierarchy. In general, an n -degree y -rough squarefree polynomial has at most

$$\left\lfloor \frac{n}{y+1} \right\rfloor$$

irreducible factors. If $R_k(F; y)$ counts the decompositions in [theorem 1.1](#) for which $F - g$ has exactly k irreducible factors, then

$$(5.2) \quad \sum_{\substack{k \geq 1 \\ k \text{ odd}}} R_k(F; y) - \sum_{\substack{k \geq 1 \\ k \text{ even}}} R_k(F; y) = \frac{p_y(n - y - 1)}{n(n-1)} q^{n-1} + O_n(q^{n-3/2}).$$

For $y \geq n/2$, only $k = 1$ is possible and $p_y(n - y - 1) = 1$, recovering the usual polynomial Goldbach main term.

6. THE HOMOLOGICAL-SIEVE INTERPRETATION

The roughness indicator admits a literal Euler-characteristic realization. Let

$$\mathcal{P}_y(q) = \{P \in \mathbb{F}_q[t] : P \text{ monic irreducible and } \deg P \leq y\}.$$

For a polynomial h , consider the augmented inclusion–exclusion complex

$$C_y^r(h) = \bigoplus_{\substack{S \subseteq \mathcal{P}_y(q), |S|=r \\ \prod_{P \in S} P | h}} \overline{\mathbb{Q}}_\ell,$$

with the usual alternating deletion maps. Its Euler characteristic is

$$\begin{aligned} \chi(C_y^\bullet(h)) &= \sum_{S \subseteq \mathcal{P}_y(q)} (-1)^{|S|} \mathbf{1}_{\{\prod_{P \in S} P | h\}} \\ (6.1) \quad &= \prod_{P \in \mathcal{P}_y(q)} (1 - \mathbf{1}_{\{P | h\}}) = \mathbf{1}_{\{d^-(h) > y\}}. \end{aligned}$$

Tensoring the coefficient with the Liouville sign produces $\lambda_q(h) \mathbf{1}_{\{d^-(h) > y\}}$.

On the squarefree configuration space, Frobenius orbits of roots are the cycles of σ_h , and the resulting coefficient function is exactly $\phi_{n,y}$ from (4.1). After extending scalars, this class function decomposes into irreducible S_n -characters and therefore gives a formal linear combination of the corresponding local systems in the rationalized Grothendieck group of the ordered-root cover. Its trivial-isotypic projection is the scalar

$$\text{Av}_{S_n}(\phi_{n,y}) = -\frac{p_y(n-y-1)}{n}.$$

This is the top-weight main term detected by the Chebotarev calculation; the remaining components are controlled by the geometric monodromy and Weil bounds in the Bender–Pollack argument.

The complex (6.1) does not improve inclusion–exclusion by itself. Its role is organizational: it identifies the signed coefficient object, separates its trivial component from the nontrivial cohomological terms, and makes the source-type extension [proposition 4.1](#) formal once joint equidistribution is available.

7. CONCLUDING REMARKS

The result has three important limitations.

- The degree n is fixed and q tends to infinity. No uniformity in n is asserted.
- The argument does not imply an integer Goldbach theorem. The finite field proof uses geometric monodromy and square-root cancellation that have no direct integer analogue.
- The threshold $q_0(n)$ is not optimized. Making it explicit would require tracking the exceptional loci and genus constants in the Bender–Pollack construction.

The arithmetic engine of the note is therefore existing work, while [proposition 2.1](#) is a short cycle-index computation. The contribution recorded here is the universal signed consequence [theorem 1.1](#), together with its Liouville,

prescribed-source, and homological-sieve formulations. The author would welcome references to any prior statement of this precise parity-bias theorem.

REFERENCES

- [1] A. O. Bender, *Decompositions into sums of two irreducibles in $\mathbb{F}_q[t]$* , C. R. Math. Acad. Sci. Paris **346** (2008), no. 17–18, 931–934.
- [2] A. O. Bender and P. Pollack, *On quantitative analogues of the Goldbach and twin prime conjectures over $\mathbb{F}_q[t]$* , preprint, 2009, [arXiv:0912.1702](#).
- [3] A. Entin, *Factorization statistics of restricted polynomial specializations over large finite fields*, Israel J. Math. **242** (2021), no. 1, 37–53.
- [4] P. Flajolet and R. Sedgewick, *Analytic Combinatorics*, Cambridge University Press, Cambridge, 2009.
- [5] O. Gorodetsky and W. Sawin, *Correlation of arithmetic functions over $\mathbb{F}_q[T]$* , Math. Ann. **376** (2020), no. 3–4, 1059–1106.
- [6] R. Petuchovas, *Asymptotic estimates for the number of permutations without short cycles*, Australas. J. Combin. **72** (2018), no. 1, 1–18.

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